

Article

Artificial Neural Network and Adaptive Neuro Fuzzy Inference System Hybridized Models in the Sustainable Integration of Language and Mathematics Skills: The Case of Singapore and Hong Kong

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Abstract: The four basic language skills, listening, speaking, reading, and writing play a crucial role in the development of an individual's skills in other disciplines. The current study aims to underpin the relationship between language skills and mathematics skills by focusing on the language and mathematics curricula of two consistently high-achieving countries, Hong Kong and Singapore, in the Program for International Student Assessment (PISA) rankings. In the current study, the convergent parallel mixed method was utilized that qualitative and quantitative data were composed together. Primarily, the outcomes of four language skills were determined in the native language teaching curricula of the two countries. The topics and themes related to four basic language skills were determined from the two mathematics curricula. The curricula were examined by document analysis from qualitative research methods. The analysis was conducted by examining the native language teaching and the mathematics curricula of both countries by the content analysis method. Later, the findings of the document analysis were used to develop machine learning models to find a possible positive relationship between language skills and the PISA scores. **Although a number of previous studies have found a reasonable relationship between language skills and mathematics skills, the current study results were contradictory to the ones performed previously in the literature, and considering the curricula no positive relationship between the language and mathematics skills was found.** The findings of the current study were further supported by the artificial neural network (ANN) and Adaptive Neuro Fuzzy Inference System (ANFIS) model performance metrics. Compared to an acceptable level of 0.80, significantly low R^2 values of 0.35 and 0.39 for the ANN and ANFIS models, respectively, indicated very little relationship between the language and mathematics skills.

Keywords: native language skill; mathematics skill; ANN; ANFIS; Singapore; Hong Kong; curricula



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1. Introduction

Language is a skill that people learn first from their family and then from the environment in which they live. It develops continually throughout an individual's life. The progress of listening, speaking, reading, and writing skills, which are included under understanding/explaining skills are called the four basic language skills, and they pertain to the development of every individual in the early childhood stage. It is meaningful to achieve a competent level of language skills for success across different disciplines, learning areas, and in the development of various types of intelligence. To develop four basic language skills and achieve success in other learning areas, the native language is the child's helper in all lessons, including foreign language skills, and it should be used systematically [1]. Among these different areas, mathematics skills are also included. Being successful in the field of mathematics depends on competence in language skills. Mathematics success relies

on the development of different linguistic skills in individuals. Application of these skills is crucial to improving lower levels of mathematical attainment.

Language is important in understanding how mathematical activities are shaped. It also helps us to understand how we relate language, teaching, and learning processes to other social interactions. Language-oriented research also deals with multilingual mathematics classes [2]. The influence of language on mathematical skills is very important and it can affect a student's mathematics achievement. Language-oriented research identifies four main areas in mathematics education [3].

There is a strong relationship between a student's mathematics skills and a student's learning processes, other social interactions, and multilingual contexts. It is also important to develop the competencies required in the mathematical applications of the language. Mathematical language is a tool used to express mathematical concepts. Using mathematical terms correctly and consciously helps students better understand mathematical concepts. Students can embody abstract concepts through mathematical language and improve their mathematical thinking skills. The effective use of native language in mathematics also increases a student's ability to use correct expressions and understand the results when solving mathematical problems. It is necessary to emphasize the meaning of mathematical language to students and encourage them to use the correct terms to express mathematical concepts [4].

In this context, language contributes to the development of intelligence by affecting thinking style and perception. In mathematics education, language is also an important tool for expressing and making sense of abstract concepts. The relationship between mathematics and language shows that mathematics is one of the languages developed by society to be able to decode concepts and relations [5].

The relationship between language and mathematics success is an important factor that affects the sustainability and continuity of education. The relationship between mathematics success and reading comprehension skills contributes to the development of both mathematical skills and language skills of students. In this context, an education system that is compatible with sustainable development goals should emphasize qualified education and language skills. Mathematics and language are one of the cornerstones of sustainable development. High-quality education has a significant role in upbringing individuals who will support sustainable development by enhancing their knowledge, skills, and social values. At this point, for sustainability in education, students should be offered opportunities to improve both math and language skills [6]. Education in all its dimensions gives these skills to students in terms of gaining basic life skills and academic success. The curricula, course materials, methods, and techniques that form the foundations of education systems should be revised by considering the latest advancements and the requirements of the period. In this direction, it is necessary to take advanced education systems as a role model and follow advancements in education globally.

The Program for International Student Assessment (PISA) is one of the educational surveys organized by the Organization for Economic Co-Operation and Development (OECD). PISA has been conducted every three years since 2000. The outcomes of PISA are steering guidance for developing education systems by following international innovations in education. In PISA, students in the 15-year-old age group representing each country are selected by a non-selective method and tested for their skills in the areas of reading, mathematics, and science. PISA has made it possible for countries to become observation reports for themselves and each other [7].

Among the consistent countries that regularly achieved success in mathematics and reading in PISA between 2012, 2015, and 2018 are Singapore and Hong Kong. The Pisa rankings for the above-mentioned countries are tabulated in Table 1.

Table 1. Mathematics and reading area rankings by country.

Country	2012		2015		2018	
	Mathematics	Reading	Mathematics	Reading	Mathematics	Reading
Singapore	2.	3.	1.	1.	2.	2.
Hong Kong	3.	3.	2.	2.	4.	4.

The year of PISA data used in the analysis part of this research is 2018. In Table 2, the 2018 PISA results are shown separately by country and mathematics–reading areas.

Table 2. Mathematics and reading scores by country in 2018.

Country	2018	
	Mathematics	Reading
Singapore	569	548
Hong Kong	551	524

The schooling rate is very high in Singapore. This is due to the fact that Singapore has opened its doors to students from many foreign countries. Singapore has divided the primary school program into mother tongue and English, social sciences and arts, science and mathematics education, life skills, and added advanced language skills in secondary school [8]. The Hong Kong Ministry of Education designed the key learning areas as Chinese, English, mathematics, science, technology, art, personal–social and humanities and physical education. There are seven basic learning goals in primary school education: awareness of responsibility, national identity, reading habits, language skills, learning skills, knowledge coverage, and healthy lifestyle (Education Bureau) [9].

Singapore and Hong Kong have regularly shown the same achievement rankings in both mathematics and reading areas. The reasons behind this stabilized success are likely to be based on the content of their curricula. In this context, the native language teaching curriculum and mathematics curriculum of these countries were investigated. The achievements of the language skills from the native language teaching programs of the countries and the achievements related to the language skills in the mathematics programs were determined. Consecutively, ANN and ANFIS analyses were performed to investigate the relationship between the results obtained from mathematics teaching programs and native language teaching programs.

ANN is a data-driven artificial neural network modeling technique that works similarly to the way human cells function. ANFIS, a hybrid model that combines the abilities of ANN and fuzzy logic theory, takes advantage of both data and knowledge-based modeling, which aids in solving complex world problems that commonly involve considerable uncertainty and non-linearity that require human-like thinking expertise. ANN and ANFIS involve developing different network architectures; however, the goal, which is to predict or estimate a dependent variable from some independent variables, is the same [10]. Particularly, in educational sciences, the use of data where limits cannot be determined with certainty in the analysis enables models to achieve more successful results. The selection of ANN and ANFIS modeling techniques in the current study was also related to this fact.

This research aims to examine the relationship between language skills and mathematics success. In this context, PISA, the program for international student assessment, has been taken as an example. Two countries that regularly achieved success in both reading skills and mathematics in the PISA exam were included in the scope of the research. Both the native language teaching and mathematics teaching programs of the mentioned countries, Singapore and Hong Kong, were examined. In addition to the data obtained after the examination, ANN and ANFIS analyses were performed. For this purpose, answers to the following questions were sought:

1. What is the relationship between language skills and mathematics success according to the data obtained from the native language teaching and mathematics teaching programs?
2. What is the relationship between language and mathematics skills using ANN and ANFIS?
3. What is the comparison of the results obtained from the ANN and ANFIS analyses?

2. Research Method

2.1. Research Sampling

In this research, the native language teaching and mathematics teaching curricula of Singapore and Hong Kong were examined. The research consists of three parts: content analysis of mathematics and native language curricula of the mentioned countries, ANN analysis, and ANFIS analysis. Native language teaching and mathematics curricula of Singapore and Hong Kong were accessed from official ministries of education.

The curricula investigated in the present study were retrieved from the following links:

Singapore English Language Syllabus 2020 (Primary 5–6 to secondary 1, 2, and 3) (https://libris.nie.edu.sg/sites/default/files/2020-01/primary_els-2020-_syllabus.pdf) (accessed on 15 December 2023).

Hong Kong English Language Education 2017 Primary 1–Secondary 6 (Primary 1–Secondary 6) (https://www.edb.gov.hk/attachment/en/curriculum-development/kla/eng-edu/Curriculum%20Document/ELE%20KLACG_2017.pdf) (accessed on 15 December 2023).

Singapore Mathematics Syllabus Primary one to six (<https://www.moe.gov.sg/primary/curriculum/syllabus>). (<https://www.moe.gov.sg/-/media/files/primary/2021-primary-mathematics-syllabus-p1-to-p6.pdf>) (accessed on 28 December 2023).

Hong Kong Mathematics Education Curriculum Guide Primary 1–Secondary 6 (<https://www.edb.gov.hk/en/curriculum-development/kla/ma/curr/index2.html>) (https://www.edb.gov.hk/attachment/en/curriculum-development/kla/ma/curr/ME_KLACG_eng_2017_12_08.pdf) (accessed on 28 December 2023).

2.2. Data Collection Method

Primarily, the outcomes of four language skills were determined in the native language teaching curricula of Singapore and Hong Kong. The topics and themes related to four basic language skills were determined from the two mathematics curricula. The curricula were examined by document analysis from qualitative research methods. Qualitative research is defined as research in which the data are collected using qualitative data collection methods such as behavior observation, interviews with participants, and document review, and a qualitative process is followed to present the findings holistically [11,12]. Document analysis covers the gradual analysis of materials based on the facts or facts that are intended to be investigated [12].

Additionally, in the current study, two different models were employed using ANN and ANFIS. The working principle of an ANN is similar to the principle of human brain processing. Each neuron receives input, processes it, and produces an output signal, which is then passed to the next layer of neurons. ANFIS, on the other hand, is an adaptive network approach that is considered a hybrid model that combines the benefits of ANN and fuzzy logic [13]. One of the significant advantages of ANFIS over the other models is that it combines the abilities of ANN and fuzzy logic to solve complex non-linear relationships between the input and output variables. The ANFIS model mixes the learning algorithms of both ANNs and fuzzy logic to overcome the problems of complex non-linearity between input and output variables [14,15].

Considering the purpose of the research, it was decided to use mixed methods in this study. We aimed to find better answers to the research aims and sub-objectives by using mostly quantitative and qualitative methods simultaneously in the mixed method

content [16,17]. The mixed methods approach is based on three models: the first is the sequential explanatory mixed method, the second is the convergent parallel mixed method, and the third is the exploratory sequential mixed method. In this study, the “convergent parallel mixed method” was used, in which qualitative and quantitative data were collected together. Quantitative and qualitative data were analyzed separately, and as a result of the findings, it was also taken into consideration whether these findings were related to each other or supported by each other [18].

2.3. Data Analysis

The native language teaching and mathematics curricula of both countries were examined by the content analysis method. Content analysis is used in social sciences and refers to the separation of content as code and theme in exchange for certain forms [12,19]. Among the learning outcomes for the curricula at each grade level, those related to language skills were examined. The themes identified for each country are shown in a tabular form.

The themes related to language skills in the mathematics program of the countries participating in the study are shown in Table 3.

Table 3. The themes in the mathematics and native language curricula of the subject countries.

Themes in the Mathematics Curricula
Working in groups
Writing/making a story/making a list
Explaining/discussing/sharing/speaking
Reading
Themes in the Native Language Curricula
Listening
Speaking
Reading
Writing

The themes and common themes obtained based on the outcomes of the mathematics curricula of the subject countries were determined. Then, we determined which of the identified themes could be included in language skills. The code numbers and frequencies obtained from this were compared with the 2018 PISA scores.

Two different combinations were created according to ANN and ANFIS in the comparison. Combinations, which mean two separate models for the ANN and ANFIS part of the research, are necessary for developing architectures [20]. Combination models tabulated in MS Office Excel spreadsheet environment were analyzed by using ANN and ANFIS tools in MATLAB version 2019a, a multi-paradigm numerical calculation software.

The development of ANN was inspired by the human brain. ANN is based on computational learning and it uses a machine learning approach to develop models by mimicking the processing of a human brain [21]. Rules are set by providing input and output information during learning in artificial neural networks. ANN commonly employs a multi-layer perceptron (MLP) model, which was also utilized in the present work. MLP consists of input, hidden, and output layers in its architecture. Other parameters involved in ANN model development are the training and activation functions, of which several options are available and can be selected by trial and error according to the model performance. According to the model performance results, the optimum performing training and activation functions were found to be the Levenberg Marquardt and hyperbolic tangent functions, respectively.

ANFIS models adopt the ANN working principles while adding the benefits of fuzzy logic systems, which make it a hybrid modeling technique. ANN parts are responsible for the data-driven systems while the fuzzy logic part is responsible for the knowledge-driven system. Although ANFIS has been previously utilized in many research areas, the research regarding the applicability of using ANFIS in educational sciences is still

vague. However, the hybrid nature of ANFIS can make it a remarkable technique in this field. ANFIS architecture consists of five distinct layers, each of which performs unique calculations to optimize the model performance. In the current study, MATLAB 2019a version ANFIS toolbox user interface was used. The membership function utilized was the Gaussian bell. The input parameters included verbal skills of teaching programs and the output parameters were PISA scores. The data splitting was conducted as follows: 70% for training, 15% for testing, and 15% for validations.

Training in ANFIS involves forward- and back-pass calculations until the model finds the optimum outcomes. The first four layers perform the forward passes by using the least-squared error method while the fifth layer performs the back-pass calculations by using the gradient descent method. The first layer in ANFIS is the input for performing the fuzzification process. In this layer, the input data points were fuzzified by using the Gaussian membership function. The second layer uses the fundamental fuzzy calculations by adopting the “AND” and “OR” functions as specified in the fuzzy logic theory. In the third and fourth layers each rule and the firing power are normalized to ensure that the rule contributions are relative. The fifth layer is where all the incoming signals from the previous layers are summed and converted to a single crisp value to be able to make meaningful evaluations.

The model performance evaluation for artificial intelligence-driven models such as the models employed in the current study was conducted by using the statistical indicator metrics. Although some metrics are available in the literature, root mean square error (RMSE) and coefficient of determination (R^2) were utilized in this study due to their popular utilization in the literature.

3. Results

The results obtained from the analysis of the data are structured according to the subheadings. The themes related to language skills in the mathematics curricula of both countries used in the ANN and ANFIS analyses are shown below.

3.1. Findings on Language Skills in Singapore Mathematics Curriculum Outcomes

The outcomes of the Singapore primary mathematics curriculum related to language skills according to grade levels are given in the following tables. As can be seen in Table 4, the largest number of themes related to language skills was included in the topic of “2DShapes”. The minimum number was found in the topic of “Time”.

Table 4. Mathematics curriculum of Singapore primary 1st-grade level by subject.

Subject	Inferences of Language Skills	Frequency
	Primary 1st Grade	
Numbers up to 100	Working in groups, describing	2
Addition and subtraction	Working in groups, writing, writing	3
Multiplication and division	Sharing, explaining, discussing, sharing	4
Money	Communicating, sharing, working in groups	3
Length	Working in groups, describing	2
Time	Telling	1
2DShapes	Describing, identifying, describing, describing, working in groups, working in groups, explaining	7
Picture Graphs	Working in groups, answering, discussing, describing, making a story	5
		27

As can be seen in Table 5, the largest number of themes related to language skills in 2nd grade were included in the topic of “3DShapes”. The minimum number was found in the topic of “Addition and subtraction”.

Table 5. Mathematics curriculum of Singapore primary 2nd-grade level by subject.

Subject	Inferences of Language Skills	Frequency
Primary 2nd Grade		
Numbers up to 100	Giving examples, talking, working in groups, counting, explaining, describing	6
Addition and subtraction	Writing, explaining, working in groups, working in groups	4
Multiplication and division	Working in groups, explaining, writing, working in groups, sharing	5
Fraction of a whole	Giving examples, using language, describing, explaining, explaining	6
Addition and subtraction	Working in groups, writing	2
Money	Reading, writing, writing, writing, working in groups, creating words	6
Length, mass, and volume	Working in groups, working in groups, explaining, describing	4
Time	Counting aloud, reading, telling, giving, writing	5
2DShapes	Describing, working in groups, explaining	3
3DShapes	Describing, describing, working in groups, working in groups, working in groups, explaining, explaining	7
Picture Graphs with scales	Working in groups, writing, explaining, making a story	4
		52

Table 6 shows that, in the primary 3rd-grade level, the theme most related to language skills was identified in the topic of “Multiplication and division”. This result was followed by the topic of “Numbers up to 10,000” with a frequency of 6. The least possible outcomes associated with language skills were found in the topics of “Addition and Subtraction” and “Angles”.

Table 6. Mathematics curriculum of Singapore primary 3rd-grade level by subject.

Subject	Inferences of Language Skills	Frequency
Primary 3rd Grade		
Numbers up to 10,000	Discussing, working in groups, counting, explaining, using language, describing	6
Addition and subtraction	Working in groups, working in groups, discussing	3
Multiplication and division	Working in groups, making stories, writing, working in groups, writing, reinforcing language, working in groups	7
Equivalent fractions	Discussing, explaining, making a list, working in groups, explaining	5
Addition and subtraction	Working in groups	1
Money	Discussing, working in groups	2
Length, mass, and volume	Counting aloud, working in groups, working in groups, working in groups	4
Time	Describing, telling, writing, working in groups	4
Area and perimeter	Explaining, working in groups, working in pairs	3
Angles	Using language	1
Perpendicular and parallel lines	Giving examples, explaining, working in pairs, working in pairs	4
Bar graphs	Working in groups, discussing, making a story, discussing	4
		44

As can be seen in Table 7, the largest number of themes related to language skills in 4th grade are included in the topics of “Numbers up to 100,000” and “Decimals up to 3 decimal places”. The minimum frequency was found (0) in the topics of “Fractions of a set of objects” and “Angles”.

As can be seen in Table 8, it has been observed that there were no differences in the frequency of the themes in the 5th grade from each other. The minimum frequency was found (1) in the topics of “Ratio”, “Angles”, and “Average of a set of data”.

Table 7. Mathematics curriculum of Singapore primary 4th-grade level by subject.

Subject	Inferences of Language Skills	Frequency
	Primary 4th Grade	
Numbers up to 100,000	Working in groups, explaining, reading, writing, describing	6
Factors and multiples	Working in groups, making a list, writing	3
Four operations	Explaining, working in groups	2
Mixed numbers and improper fractions	Giving examples	1
Fractions of a set of objects	-	0
Addition and subtraction	Working in groups	1
Decimals up to 3 decimal places	Counting, writing, describing, explaining, writing, determining	6
Addition and subtraction	Working in groups, explaining	2
Multiplication and division	Working in groups	1
Time	Writing, reading, writing, describing, working in groups	5
Area and perimeter	Making a figure	1
Angles	-	0
Rectangle and square	Describing, discussing, and working in pairs	3
Line symmetry	Working in groups, working in pairs	2
Tables and line graphs	Explaining, discussing, explaining, discussing	4
		37

Table 8. Mathematics curriculum of Singapore primary 5th-grade level by subject.

Subject	Inferences of Language Skills	Frequency
	Primary 5th Grade	
Numbers up to 10 million	Discussing, reading, writing	3
Four operations	Explaining, sharing ideas	2
Fraction and division	Writing, explaining	2
Four operations	Discussing, working in groups	2
Percentage	Talking, working in pairs, working in groups	3
Ratio	Working in groups	1
Rate	Talking about, talking	2
Area of triangle	Working in groups, explaining	2
Volume of cube and cuboid	Working in groups, determining	2
Angles	Working in groups, working in pairs	2
Triangle	Explaining, describing, explaining	3
Angles	Describing	1
Parallelogram	Discussing, explaining	2
Average of a set of data	Discussing	1
		28

In Table 8, the largest number of themes related to language skills in 6th grade are included in the topic of “Area and circumference of the circle”. The minimum frequency was found (0) in the topics of “Ratio”, “Algebra”, and “Volume of cube and cuboid”.

As it can be deduced in Table 9, the outcomes of the Singapore primary mathematics curriculum related to language skills at all grade levels, the main focus was observed in the themes of “describing, identifying, working in groups, and explaining”.

The outcomes of the Hong Kong primary mathematics curriculum related to language skills according to grade levels are given in the following tables. In Table 10, the largest number of themes related to language skills was included in the topic of “Shape and Space”. There were no outcomes related to language skills in other topics.

Table 9. Mathematics curriculum of Singapore primary 6th-grade level by subject.

Subject	Inferences of Language Skills	Frequency
Four operations	Primary 6th Grade	
Percentage	Working in groups	1
Ratio	Giving examples, explaining	2
Distance, time, and speed	-	0
Algebra	Talking about, discussing, talking about	3
Area and circumference of the circle	-	0
Volume of cube and cuboid	Describing, working in pairs, working in groups, working in groups	4
Special quadrilaterals	-	0
Nets	Explaining	1
Pie charts	Working in groups, discussing	2
	Discussing, reading	2
		14

Table 10. Mathematics curriculum of Hong Kong primary 1st- to 3rd-grade levels by subject.

Subject	Inferences of Language Skills	Frequency
	Primary 1st to 3rd Grades	
Number	-	0
Measures	-	0
Shape and Space	Describe, describe	2
Data handling	-	0
		2

In Table 11, there are no outcomes related to language skills in any subject of the mathematics curriculum at the 4th- to 6th-grade levels.

Table 11. Mathematics curriculum of Hong Kong primary 4th- to 6th-grade levels by subject.

Subject	Inferences of Language Skills	Frequency
	Primary 4th to 6th Grades	
Number	-	0
Algebra	-	0
Measures	-	0
Shape and space	-	0
Data handling	-	0
		0

As can be seen in Table 12, outcomes were determined with language skills, which were different from other levels. Higher numbers of outcomes related to language skills were found in the topics of “Number and Algebra” and “Measures, shape, and space”.

Table 12. Mathematics curriculum of Hong Kong junior grade level by subject.

Subject	Inferences of Language Skills	Frequency
	Primary junior secondary	
Number and algebra	Describe	1
Measures, shape, and space	Describe	1
Data handling	Describe	1
Number and algebra	Junior senior secondary, describe	2
Measures, shape, and space	Describe, communicate	2
Data handling	Describe	1
		8

When comparing the Hong Kong mathematics curriculum with the Singapore mathematics curriculum, the themes related to language skills were identified the most in Singapore among mathematics outcomes.

3.2. Findings on the Themes and Number of Outcomes in the Native Language Teaching and Mathematics Programs of the Subject Countries

In this section, the content of the country programs is detailed. The basic language skills included in the native language curriculum are shown. Themes related to language skills were created in mathematics teaching programs.

The number of outcomes in each language skill in the native language curriculum of the subject countries was tabulated. In the same way, the number of mathematics outcomes related to language skills themes in mathematics teaching programs is also shown.

The outcome numbers of each language skill are shown in Table 13. Hong Kong has the highest number of outcomes in listening, reading, and writing skills, while Singapore has the highest number of outcomes in speaking skills. Singapore, which also showed regular success in the field of reading in the PISA exam, showed the least number of outcomes in reading skills in the native language curriculum.

Table 13. The skills and the number of outcomes in the native language curricula of the subject countries.

Skills	Country	
	Singapore	Hong Kong
Listening	10	17
Speaking	17	15
Reading	9	28
Writing	13	17

Table 14 shows the number of themes related to language skills obtained from countries' mathematics programs. In this context, the Singapore mathematics program differed significantly from Hong Kong.

Table 14. The themes and the number of outcomes in the mathematics curricula of the subject countries.

Themes	Country	
	Singapore	Hong Kong
Working in groups	57	0
Writing/making a story/making a list	27	0
Explaining/discussing/sharing/speaking	108	10
Reading	4	0

In Figures 1 and 2 below, the weights of the language skills obtained from the above tables were graphically illustrated in a pie chart for Singapore and Hong Kong, respectively.

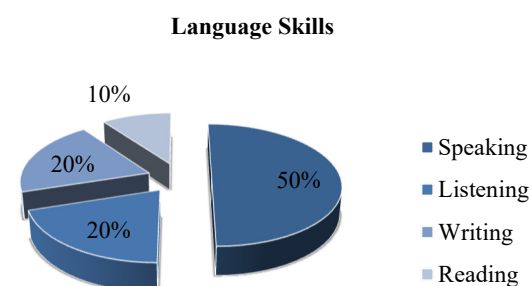


Figure 1. Four basic language skills pie charts based on Singapore math program outputs.

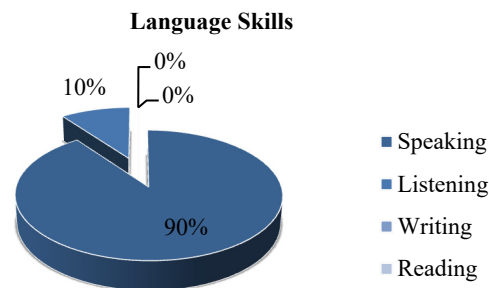


Figure 2. Four basic language skills pie charts based on Hong Kong math program outputs.

3.3. Analysis of Data Obtained from Mathematics Teaching Programs Using ANN

The ANN analysis was performed after examining the mathematics teaching programs determined within the scope of the native language education system used in both Hong Kong and Singapore. The results are illustrated in Figure 3 below. Accordingly, one individual model was developed by ANN. In the model, the relationship between verbal skills and language skills based on the PISA scores was examined. The root mean square error (RMSE) and the coefficient of determination (R^2) for the developed model were tabulated and are shown in Table 15.

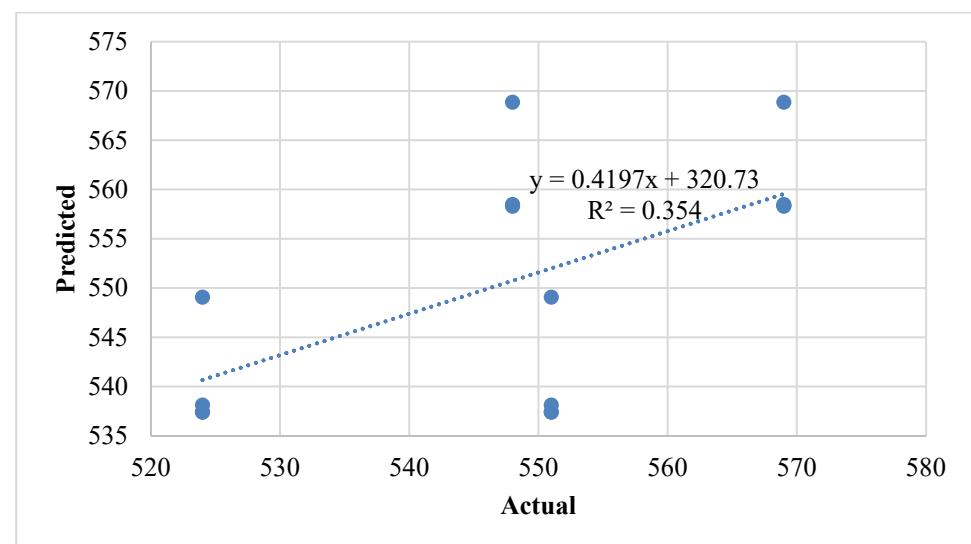


Figure 3. Relationship between the language themes in mathematics programs and language skills in native language teaching programs of the subject countries based on the PISA scores for model 1.

Table 15. RMSE and R^2 values of Model 1.

Model Number	RMSE	R^2
Model 1	17.6385	0.353954

As can be seen in Figure 3 for Model 1, the R^2 was 0.35 in the relationship between verbal skills and language skills based on the PISA scores of the subject countries. An insufficient relationship between the variables was observed (linguistic value equivalents of statistical values: 40: insufficient; 55–70: average; 70–80: very good; and 80 and above: superior significance) [22,23]. The RMSE value was 17.6385.

3.4. Analysis of Data Obtained from Mathematics Teaching Programs with ANFIS

The results of the ANFIS analysis using the data obtained from the native language and mathematics teaching programs of the subject countries covered within the scope of the research are illustrated in Figure 4 below. Accordingly, one individual model was

developed using ANFIS. In the model, the relationship between verbal skills and language skills based on the PISA scores was examined. The root mean square error (RMSE) and the coefficient of determination (R^2) for the models developed were tabulated and are shown in Table 16.

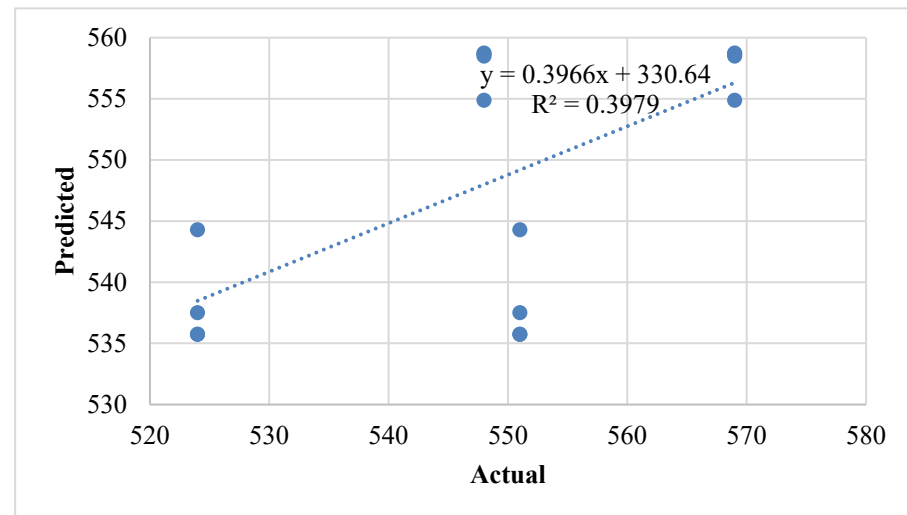


Figure 4. Relationship between the language themes in mathematics programs and language skills in native language teaching programs based on the PISA scores of the subject countries for model 2.

Table 16. RMSE and R^2 values of Model 2.

Model Number	RMSE	R^2
Model 2	12.42771	0.397867

As can be seen in Figure 4 for Model 2, the R^2 was 0.39 in the relationship between verbal skills and language skills based on the PISA scores of the subject countries. An insufficient relationship was found between them. The RMSE value was 12.42771.

The relationship between input and output is also given in Figure 5.

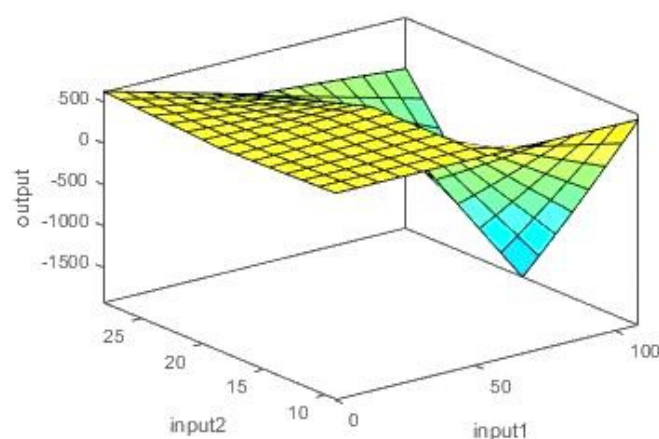


Figure 5. The relationship between input and output.

3.5. The Findings of the Comparison of ANN and ANFIS Analyses

In this section, the results obtained from the ANN and ANFIS analyses were compared. The results were tabulated in Table 17. Both analysis results indicated that there was an insufficient relationship.

Table 17. Comparison of RMSE and R^2 values in the models.

Model Number	RMSE	R^2
ANN Model 1	17.6385	0.353954
ANFIS Model 2	12.42771	0.397867

These models, developed in light of data obtained from both the native language and mathematics teaching programs of the countries in question, have shown that the relationship between language and mathematics skills is not only about the teaching programs/curricula.

4. Discussion and Conclusions

In the models created for use in the ANN and ANFIS analyses, the number of learning outcomes and codes obtained from the native language teaching and mathematics curricula of both countries are shown in tabular form. When the content analyses in the native language and mathematics teaching programs of the countries covered by the research were examined, as well as the ANN and ANFIS analyses, the following results answered the three research questions of this study:

1. What is the relationship between language skills and mathematics success according to the data obtained from the native language teaching and mathematics teaching programs?

It has been proven in many studies that there is a relationship between language skills and mathematics [24–29]. Students' competence in using their native languages is proportional to their success in mathematics. Despite this, as a result of this study, it has been found that this ratio is associated with educational programs in only one country. Textbooks, teacher activities, teaching materials, and individual differences between students in educational environments outside of educational programs may play a larger role in this relationship.

Compared to the English curriculum and the mathematics curriculum of Singapore, a large number of achievements that can be matched with four language skills in the mathematics curriculum program content were identified. The maximum number of achievements that can be related to language skills in the mathematics curriculum is in grade 2 (f 52). The wider range of subjects compared to the first year shows that the content of the program is concentrated in all respects in the second year [30]. The minimum number of achievements is in grade 6 with a frequency value of 14.

Compared to the Hong Kong English curriculum and the mathematics curriculum, no results parallel to the relationship found in the Singapore example could be reached. Frequency values were found to be "2" at grade levels of 1–3, "0" at grade levels of 4–6, and "8" at the junior grade level in the Hong Kong mathematics program. No relationship between mathematics and native language teaching programs was determined.

According to the data obtained from the content of the English language teaching programs of the subject countries, Hong Kong has the highest number of achievements compared to Singapore. It was determined that the highest number of achievements in the Hong Kong English curriculum is in reading skills. The lowest number of achievements detected in the programs is in the area of reading in Singapore (f 9).

Educational curricula have a very important place under the roof of education. In the research, Saravanan [31] reports how the integration of the mathematics curriculum and the content areas of mathematics through English have changed approaches to teaching and learning. Wong et al. [32] stated that children's proficiency or inadequacies in mathematics come immediately after English and that studies should be carried out on the mathematics curriculum. Some studies argue that student success has nothing to do solely with the curriculum. Coleman [33] has revealed that student success depends more on family and socio-economic background than on school resources. Hattie [34] has shown that teacher

effectiveness has a significant impact on student achievement, but the curriculum accounts for a much smaller part of this impact.

2. What is the relationship between language and mathematics skills using ANN and ANFIS?

The results obtained for the ANN model demonstrated R^2 values as low as 0.35, which indicated that the relationship between verbal skills and language skills based on the PISA scores was not significant. On this basis, it can be noted that the content of the mathematics teaching programs of the subject countries and the language skills gains contained in the content do not have significant effects according to the ANN model.

Although the ANFIS model results demonstrated a slightly higher R^2 value of 0.39, this outcome is still considered too low to determine the relationship between verbal skills and language skills based on the PISA scores.

In addition to the fact that there are studies in which ANN and ANFIS are used in educational sciences, there are also studies of these models included in literature reviews, especially those referenced in this research. The research of Okewu et al. [35] shows that ANN can be applied to educational sciences for analyzing educational data. Kalaycı [36] and Kalaycı Alas [37] analyzed the data obtained from the native language teaching programs of nine countries, including Hong Kong and Singapore, with ANFIS.

3. What is the comparison of the results obtained from the ANN and ANFIS analyses?

These models, developed in the light of data obtained from both native language and mathematics teaching programs of the countries in question, are two systems rather than a uniform method. Both ANN and ANFIS results have shown that the relationship between language and mathematics skills is not purely related to teaching programs/curricula. This conclusion is drawn from the fact shown in the literature that higher R^2 values close to 1 and lower RMSE scores indicate a better model performance. Since both of the model performance metrics were far below the above-mentioned target values, both models were found to be insufficient in terms of finding the relationship between language skills and mathematics.

According to statistical indicator metrics, although both models were not able to explain the relationship between verbal skills and language skills based on the PISA scores, the ANFIS model slightly outperformed the ANN model. This can be justified by the fact that ANFIS models utilize a hybrid modeling technique that combines the advantageous sides of artificial neural networks and fuzzy logic theory.

A number of previously published research works that had similar objectives but utilized different methodologies to the current study can be found in the literature. Research conducted by Cheung [38] examined the PISA 2012 Mathematics exam results in East Asian countries such as Singapore and Hong Kong and compared the mathematical competencies of disadvantaged but successful students with advantaged but low-achieving students. The study found that disadvantaged students show higher familiarity with mathematical concepts, experience less math anxiety, and have a higher math self-concept. Another study suggests that speaking a minority language at home has a positive effect on mathematics and reading achievement [39]. On the contrary, it has also been stated that being an immigrant and having a low socio-economic status usually negatively affects mathematics achievement [40]. A study conducted by Cui et al. [41] found that parents' participation in education in early childhood and positive attitudes towards education positively affect children's mathematics achievement by increasing their interest in learning. Another study conducted in South Africa showed that students who speak African languages and English well are more successful in mathematics and have an advantage over students who speak other languages [42]. In addition, it has been stated that factors such as socio-economic status, language spoken at home, and perceptions of the importance of mathematics also contribute to mathematics failure.

Research conducted in Hong Kong showed that kindergarten students of low socio-economic status experience difficulties in early reading and language skills, which nega-

tively affects their mathematics achievement [43]. English Chinese reading and mathematics skills were examined by Inoue et al. [44] and they found that the ability to understand the structural features of language (morphological awareness) had a significant effect on the overall academic achievement and mathematics performance of children in Hong Kong.

In another study, the influence of switching from Malay to English language education on mathematical success was investigated [45]. It has been observed that the mathematics performance of students declined due to escalated anxiety because of the language change and, as a result, the Malay language was re-adopted in the teaching program. This situation emphasized the importance of education in the mother tongue. Lu et al. [46] also stated that the phonological structure of Chinese is related to mathematics achievement, and students who speak Chinese are more successful.

There are also studies that concluded that close relationships between parent–child and teacher–child positively affect children’s mathematical abilities through language skills [47]. In addition, it was concluded that education in the mother tongue is more effective in mathematics achievement [48–51].

The relationship between language and mathematics achievement cannot be determined by examining the teaching programs alone. The relationship between language and mathematics can be measured by all the elements of a country’s education system and individual differences. Many factors such as classroom programs, curricula, teacher training programs, classroom activities, individual intelligence areas, textbooks, and measurement and evaluation applications are effective in the relationship. In this study, an attempt was made to explore how applications such as ANN and ANFIS can be applied in educational sciences based on teaching programs. Although a number of studies found in the literature have shown that AI techniques can successfully be implemented in education sciences, the poor performance of the model outcomes in the current study may be linked to the lack of amount of data points used [52–54]. Additionally, **integrated data learning models such as ensemble learning could have achieved better performance scores. Considering the above-mentioned limitations, the authors suggest improvements for future research.**

It is also noteworthy to mention that the current study solely focuses on the relationship between language and mathematics skills, which could limit the ability of the models to identify the pattern in the data. However, including additional skills in the models may have improved the model’s performance. Additionally, only two countries’ curricula were investigated and modeled due to their consistency in achieving high PISA scores; however, increasing the number of countries and adding their curriculum in consideration may have a positive influence on the model performance results. The above-mentioned limitations of the current study may have a place in future research as room for development in the field of education and artificial intelligence. **The authors strongly encourage future research regarding the application of machine learning tools such as ANN and ANFIS, which were utilized in the current study to ensure the sustainable development of educational programs and curricula.** The authors also believe that this pursuit would aid in achieving the United Nations Sustainable Development Goal 4.

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